INCH-POUND

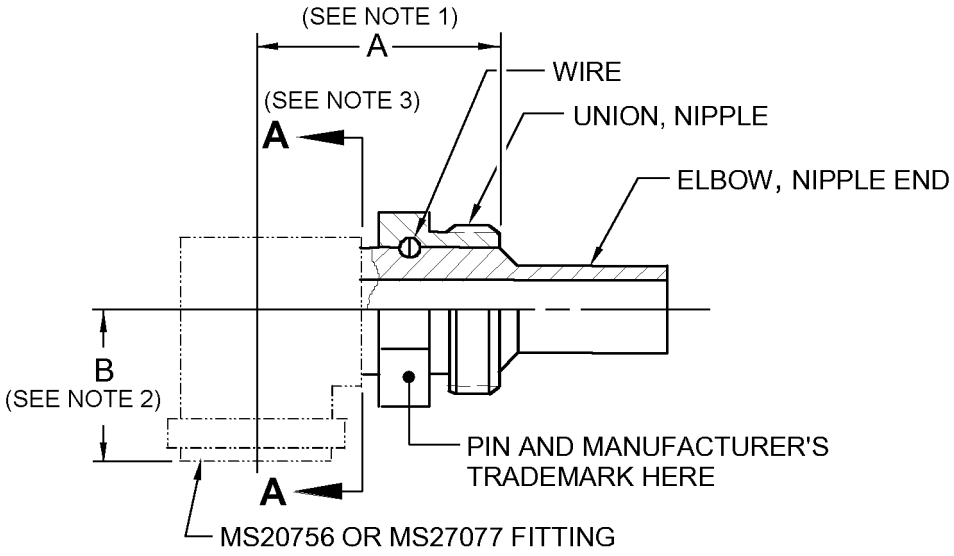
MS27066E
24 September 2003
SUPERSEDING
MS27066D
30 June 1972

DETAIL SPECIFICATION SHEET

ELBOW SUBASSEMBLY, SWIVEL FLANGE, 90°

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-DTL-27272.



NOTES:
1. Dimension A is measured between the hose end of the threaded section of the union and the centerline of the insert.
2. Dimension B is measured between the centerline of the nipple and the free end of the insert.
3. Any bent tube design of elbow subassembly components to the left of plane A-A is acceptable provided that the requirements of this specification sheet and the procurement specification are met.

FIGURE 1. Elbow subassembly illustration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T15:15Z
Check the source to verify that this is the current version before use.

REQUIREMENTS

Intended use. This part is a component of MS27058.

Identification of product. The Part or Identifying Number (PIN) and the manufacturer's trademark shall be permanently marked on the assembly. The PIN for this assembly shall be as specified in table I (e.g., MS27066-8C).

Dimensions. Dimensions are in inches.

Material. PIN suffix C, corrosion-resistant steel.
PIN suffix D. Aluminum alloy.
No PIN suffix. Combination of aluminum alloy and corrosion-resistant steel.
See applicable specification sheet for other components identified in table I for their material requirements.

Finish. Corrosion-resistant steel, do not passivate.
Aluminum alloy. Anodize in accordance with MIL-A-8625, type II, dye blue.

Elbow subassembly illustration. See figure 1.

Order of precedence. This specification takes precedence over the documents referenced herein. Unless otherwise specified, referenced documents shall be of the issue in effect on the date of solicitation.

TABLE I. Elbow subassembly composition.

PIN	Wire PIN	Union, nipple PIN	Elbow, nipple end PIN	A 1/ ±.035	B 2/ ±.035
MS27066	MS27072	MS27071	MS27082		
-8C	-8C	-8C	-8C	1.135	.892
-8			-8D		
-10C	-10C	-10C	-10C	1.315	.896
-10		-10D	-10D		
-12C	-12C	-12C	-12C	1.808	1.156
-12		-12D	-12D		
-16C	-16C	-16C	-16C	1.901	1.282
-16		-16D	-16D		
-20C	-20C	-20C	-20C	2.180	1.500
-20		-20D	-20D		
-24C	-24C	-24C	-24C	2.500	1.688
-24		-24D	-24D		

1/ Dimension A is measured between the hose end of the threaded section of the union and the centerline of the insert.
2/ Dimension B is measured between the centerline of the nipple and the free end of the insert.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T15:15Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 99
DLA - CC

Review activities:
Army - AR, AT, MI
Navy - MC, SA, SH
Air Force - 71

Preparing activity:
DLA - CC

(Project 4730-0868-037)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T15:15Z
Check the source to verify that this is the current version before use.